



Beyond cichlid diversification: environmental drivers of biodiversity across the Nicaraguan lake system

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Abstract The Nicaraguan lake system, comprising two Great Lakes and several isolated crater lakes, is a natural laboratory for investigating how environmental heterogeneity and isolation shape biodiversity and a classic model for Midas cichlid diversification. We asked whether lake characteristics associated with cichlid diversification are also linked to prokaryotic and eukaryotic diversity patterns. Using environmental DNA metabarcoding of water and sediments, we characterized communities in the large, shallow Great Lakes and smaller, deeper crater lakes. Lakes differed markedly in environmental conditions, and community composition differed between water and

sediments, while patterns suggested higher prokaryotic diversity in sediments and greater eukaryotic diversity in the water column; however, sediment samples were not replicated. At sampled sites, the Great Lakes harbored highly diverse communities, potentially reflecting their size and hydrological connectivity, whereas crater-lake communities showed stronger differentiation. Mean lake depth was strongly associated with community composition, particularly in eukaryotes. Thus, a lake characteristic previously implicated in Midas cichlid diversification is also associated with broader biological community structure. Overall, our results provide partial support for shared environmental structuring, but relationships differed among taxonomic groups and did not necessarily parallel patterns of cichlid diversification. These findings highlight both common and taxon-specific responses to lake environmental variation.

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Introduction

Biodiversity is unevenly distributed across the globe. Understanding why some regions harbor more species than others is a fundamental challenge in ecology and biogeography. Several broad biodiversity patterns have been identified, including latitudinal

gradients, with richness generally increasing toward the tropics; productivity–diversity relationships, with systems of greater available energy often supporting more species; habitat heterogeneity, under which more complex environments support higher species diversity; and species–area relationships, with larger areas tending to harbor more species (MacArthur & MacArthur, 1961; Pianka, 1966). Contemporary patterns of biodiversity are also shaped by historical geological, climatic, and evolutionary processes (Gaston, 2000; Wiens & Donoghue, 2004). Island biogeography theory (MacArthur & Wilson, 1963, 1967) has been instrumental in explaining these patterns by underlining the roles of area and isolation in shaping colonization, extinction, and diversification. This theory highlights that isolation can both constrain colonization, thereby reducing species richness, and promote diversification by restricting gene flow and favoring endemism and adaptive radiation (Whittaker & Fernández-Palacios, 2006; Field & Ricklefs, 2016). Archipelagos, both oceanic and terrestrial (those formed by freshwater lakes), are crucial hotspots of global biodiversity, providing valuable natural laboratories for examining how area, isolation, environmental heterogeneity, and ecological opportunity interact to shape biodiversity.

Tropical freshwater ecosystems are hotspots of global diversity, supporting a disproportionately high number of species (Dudgeon et al., 2006; Sayer et al., 2025). The combination of high primary production, climatic stability, and spatial heterogeneity favors an exceptional concentration of micro-endemic species, and the fine resource partitioning allows for the coexistence of species (Albert & Crampton, 2010; Magurran et al., 2025). Tropical lakes are especially noteworthy, acting as evolutionary ‘islands,’ where isolation and intense ecological interactions have driven explosive speciation, resulting in some of the highest concentrations of unique species on Earth (Seehausen, 2006). Biodiversity in these regions, however, remains incompletely documented, particularly for invertebrates and microbes (Dudgeon et al., 2006; Sundar et al., 2020; Schoenle et al., 2024), preventing proper estimates of species richness. Even vertebrates, including comparatively well-studied groups such as fishes, persist poorly characterized (Snoeks, 2000; Stiassny, 2002; Chapman et al., 2008). In this context, environmental DNA (eDNA) approaches provide a powerful tool for surveying

neglected biodiversity in these typically understudied systems. Environmental DNA provides potentially complementary temporal and spatial views of biodiversity, where water might capture a contemporary snapshot of present species, while sediment preserves a more stable, long-term record of the ecosystem’s biological history (Harper et al., 2019; Ataman et al., 2025; Iacaruso et al., 2026).

In the Neotropics, some of the major lacustrine systems occur along the Pacific coast of Nicaragua, including the two Great Nicaraguan lakes (L), L Nicaragua (Cocibolca) and L Managua (Xolotlán), and several small volcanic crater lakes (CL). This region lies within the Central American Volcanic Arc, one of the most active volcanic regions in the world (Kutterolf et al., 2007). The Great Nicaraguan lakes are large (8143 and 1053 km², respectively), and relatively old, dating from the early Pleistocene (Bussing, 1976). The chain of crater lakes is much younger, dating from the Holocene (Kutterolf et al., 2007). The Great Nicaraguan lakes were connected to each other in the past (Villa, 1976) and are still occasionally connected. These lakes are associated with an array of inflow rivers and drain through the San Juan River into the Atlantic Ocean. The Great Nicaraguan lakes are relatively shallow (mean depth 9 m; Bussing, 1976) and exposed to strong wind-induced sediment resuspension, resulting in overall turbid waters (Vammen et al., 2006). In contrast, the crater lakes are completely isolated, considerably smaller (from 1 to 21 km²), and much deeper, with depths of up to 200 m. Due to their great depth and high crater rims, crater lakes are more protected from the wind and generally exhibit clearer waters (Table 1). Research in this region has historically focused primarily on fish communities (Thorson, 1976b, a), leaving a knowledge gap regarding other aquatic organisms. More comprehensive surveys across the region are available for mollusks (Pérez & López, 2002, 2003), whereas studies on phytoplankton (Hooker et al., 1991), macrophytes (Cole, 1976), and benthic invertebrates (Bijlmakers & Sobalvarro, 1988; Ráudez Reyes et al., 2021) have mostly been limited to individual lakes.

Evidence from fish studies indicates that the two Great Nicaraguan lakes harbor high species richness, with more than 40 species recorded in L Nicaragua, whereas the crater lakes are considerably less diverse, with only four fish species recorded in CL Asososca León, the most isolated crater lake (Waid

Table 1 Characteristics of the Nicaraguan lakes

	L Nicaragua	L Managua	CL Masaya	CL Xiloá	CL Apoyo	CL Asososca León
Age* (Years)	500,000	500,000	6,000	6,100	23,890	4,500
Area* (km ²)	8143	1053	8.38	3.75	21.10	0.81
Mean depth* (m)	13	7.80	41.70	60	142	17.2
No. fish species**	64	28	11	18	14	4
Temperature (°C)	28.04	27.50	29.81	29.53	29.65	30.66
Dissolved oxygen(mg/l)	5.00	4.50	6.20	4.90	6.90	6.70
Dissolved carbon(mg/l)	3.94	8.40	2.02	2.41	2.30	2.63
Nitrates (mg/l)	0.62	0.82	0.30	0.24	0.17	0.27
Phosphates (mg/l)	0.042	0.478	0.01	0.23	0.02	0.02
Salinity (PSU)	0.11	1.11	0.20	5.16	3.43	1.16

*Data taken from Waid et al. (1999). **Data taken from Angulo et al. (2023)

et al., 1999; Angulo et al., 2023). The high species richness of the large lakes may be explained by their greater nutrient availability, together with their larger size, hydrological connectivity, and habitat heterogeneity, all of which likely increase resource availability (Härer et al., 2021). In contrast, crater lakes are more isolated, oligotrophic, and deeper, and although they support fewer species, they exhibit higher levels of endemism (Bussing, 1976; McKaye et al., 2002; Stauffer et al., 2008; Santacruz et al., 2020). Cichlids constitute a dominant component of the fish fauna in the Nicaraguan lakes, and the Midas cichlid species complex (*Amphilophus* spp.) accounts for the largest proportion of fish biomass. Extensive research on this species complex has shown that the Great Nicaraguan lakes served as source populations for the repeated and independent colonization of the crater lakes, where populations rapidly adapted and diversified (Barluenga et al., 2006; Barluenga & Meyer, 2010; Kautt et al., 2016, 2020). In the Great Lakes, cichlid diversification is associated with adaptation to contrasting sandy and rocky substrates, accompanied by the evolution of hypertrophied lips (Barlow & Munsey, 1976; Machado-Schiaffino et al., 2014; Sowersby et al., 2021). In the crater lakes, by contrast, the major axis of diversification is linked to a new ecological opportunity: the colonization of the open water column. In deep crater lakes, this process has repeatedly produced high-bodied benthic and elongated limnetic ecotypes (Barlow & Munsey, 1976; Barluenga et al., 2006; Elmer & Meyer, 2011; Kusche et al., 2014;

Leal-Cardín et al., 2024). Despite sharing many features, the crater lakes differ in their degree of isolation and other ecological characteristics, suggesting that their processes shaping biodiversity may also vary among lakes (Dittmann et al., 2012).

A previous eDNA metabarcoding survey of the Nicaraguan lakes characterized prokaryotic and eukaryotic communities in water samples and showed that community composition varied markedly among lakes, with the environmental heterogeneity emerging as a major correlate of biodiversity patterns (Härer et al., 2021). That study also reported generally higher diversity in the Great Nicaraguan lakes than in the crater lakes and identified the dominant taxonomic groups across the lake system. However, it focused on shoreline water samples and broad regional patterns, without evaluating vertical variation within lakes or comparing pelagic and benthic habitats. Moreover, it remains largely unexplored whether lake characteristics associated with Midas cichlid diversification are also related to broader patterns of biodiversity across the lake system. Addressing this question requires an integrated characterization of biological communities and environmental variation across lakes.

Here, we aimed to identify the main biodiversity patterns in the Nicaraguan lake system by characterizing prokaryotic and eukaryotic communities in water and sediments, comparing these patterns with fish richness, and analyzing their associations with abiotic lake factors. Specifically, we investigated whether lake characteristics previously associated

with Midas cichlid diversification were also linked to broader patterns of prokaryotic and eukaryotic diversity and community composition. To this end, we (1) characterized the environmental variation among the Nicaraguan lakes by compiling information on abiotic lake factors, (2) estimated diversity patterns of prokaryotic and eukaryotic communities in water and sediments using environmental DNA (eDNA) meta-barcoding, and (3) assessed associations between biodiversity patterns and physical and chemical factors. Together, these analyses contribute to an ecological baseline for the Nicaraguan lake system and allow us to assess the extent to which environmental gradients associated with Midas cichlid diversification overlap with those structuring broader prokaryotic and eukaryotic communities.

Materials and methods

Physical and chemical features of the Nicaraguan lakes and environmental sample collection

We studied six lakes in Nicaragua, including the two Great Lakes (L), L Nicaragua and L Managua, and four nearby crater lakes (CL), CL Apoyo, CL Asososca León, CL Masaya, and CL Xiloá, over two field campaigns performed in November in 2019 and 2021 (Fig. 1). Fieldwork coincided with the end of the rainy season and the main breeding period of fish. All measures and samples were taken at approximately the same time of the day, 8 am, to avoid temporal biases. We measured physical and chemical parameters of surface water at one site per lake at a depth of 1 m using a multiparameter probe (YSI ProDSS): temperature, pH, dissolved oxygen, concentration of nitrate, phosphate, dissolved organic carbon (DOC), total dissolved solids (TDS), conductivity, and salinity. We also compiled data from the literature on lake

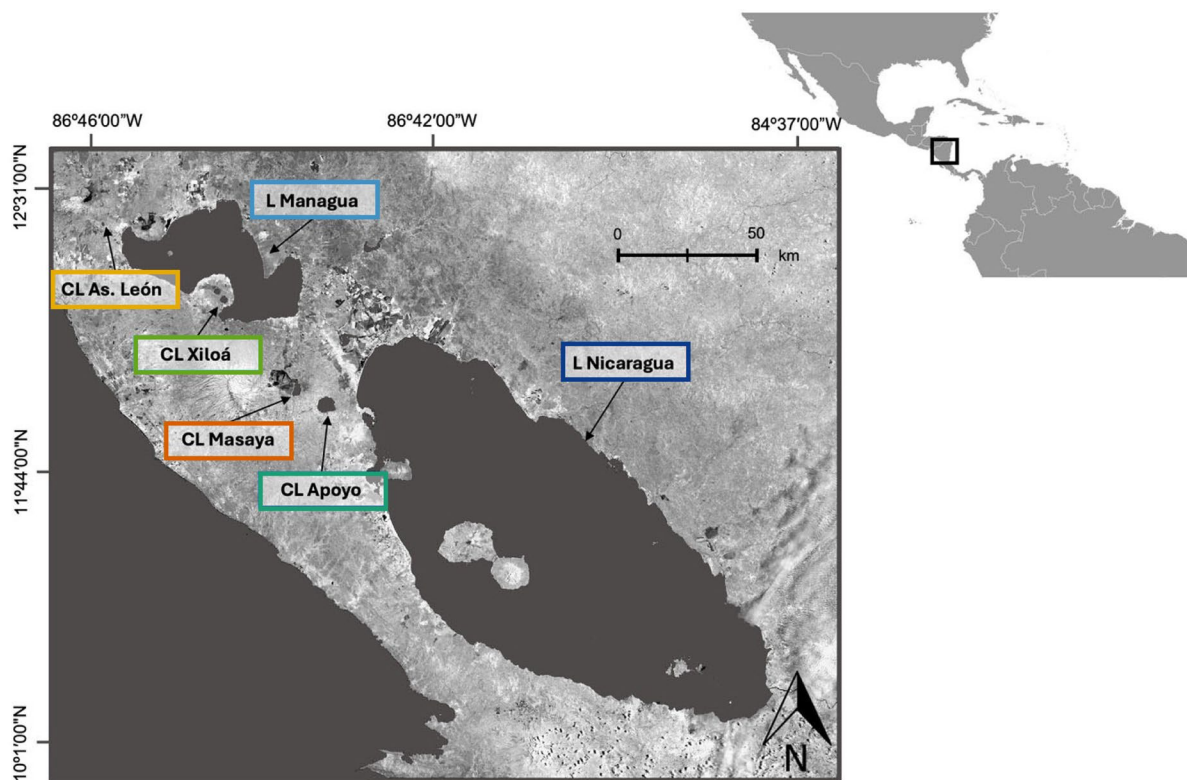


Fig. 1 Map of the study area showing an overview of the Pacific coast of Nicaragua with the Great Nicaraguan Lakes and the crater lakes

physical variables, including area, maximum and mean depth, and age.

Environmental samples, including sediments and water collected at different depths, were obtained from a single site in each lake to capture broad inter-lake differences and, where possible, vertical variation within lakes. Accordingly, the sampling design was intended to characterize broad differences among lakes rather than horizontal spatial heterogeneity within lakes. The design was unbalanced because sampling depth and replication differed among lakes and sampling campaigns owing to logistical constraints and the morphology of the lakes, particularly the large, shallow Great Lakes, where deep-water habitats are scarce and difficult to access. Water samples were collected during two sampling campaigns. In 2019, a single surface and a single deep-water sample were collected from each lake. Samples were collected at 0 and 10 m in L Nicaragua and at 0 and 20 m in the crater lakes. Lake Managua and CL Masaya were not sampled in 2019. These samples were analyzed only for eukaryotic diversity (COI metabarcoding). In 2021, the sampling design incorporated field replication and sediment sampling. Sediment samples were collected at 5 m in all lakes except L Managua, where no sediment sample could be retrieved. In the crater lakes Asososca León, Xiloá, and Apoyo, two replicate water samples were collected at 0 m and two at 10 m. In CL Masaya, one water sample was collected at 0 m and two at 10 m. In L Nicaragua, only two replicate surface water samples (0 m) could be collected because no deep-water sample could be obtained. In L Managua, two replicate surface water samples (0 m) and one water sample at 5 m, the maximum accessible depth, were collected. Samples collected in 2021 were analyzed for both prokaryotic (16S metabarcoding) and eukaryotic (COI metabarcoding) analyses. We collected a total of 33 samples, comprising 28 independent water samples and five sediment samples (Table S1).

Water samples were collected using a 5-l Niskin bottle attached to rope. At each sampling point, 500 ml of water was collected and filtered through 47-mm-diameter cellulose filters with a 0.2- μ m pore size using a manual vacuum pump. Filters were stored in Qiagen ATL lysis buffer, which preserves DNA integrity (Majaneva et al., 2018), and kept at -20°C until DNA extraction. Sediment samples were directly collected in sterile 50 ml tubes, stored

in ethanol, and preserved at -20°C until DNA extraction. Additionally, we compiled information on lake physical features and fish diversity from the literature for each lake (Table 1; Table S2).

DNA extraction, amplification, library preparation, and sequencing

DNA was extracted from water filters using the QIAGEN DNeasy Blood & Tissue Kit (Qiagen) and from sediment samples using the QIAGEN DNeasy PowerSoil Pro Kit (Qiagen). Filters were cut into small pieces using sterile instruments and incubated 56°C overnight with 20 μ l proteinase K and 180 μ l ATL buffer to promote lysis and DNA release from the filter, centrifuged, and the supernatant was used for DNA extraction following the manufacturer's protocol. Prior to DNA extraction, ethanol-preserved sediment samples were allowed to settle or briefly centrifuged, and the ethanol was carefully removed. The sediment was gently homogenized, and approximately 250 mg was transferred to a PowerBead Pro Tube for DNA extraction using the DNeasy PowerSoil Pro Kit. After extraction, DNA concentration was quantified using a Qubit 4.0 Fluorometer (Life Technologies). To target the prokaryotic community, we amplified a 253 bp fragment of the V4 region of the 16S rRNA gene using the forward primer F515 (GTGYCAGCMGCCGCGGTAA) and the reverse primer R806 (GGACTACNVGGGTWTCTAAT) (Apprill et al., 2015; Parada et al., 2016). To characterize the eukaryotic community, we amplified a 313 bp fragment of the Cytochrome Oxidase I gene (COI) using the forward primer mlCOIintF (GGWACWGGWTGAACWGTWTAYCCYCC) and the reverse primer jgHCO2198 (TAIACYTCIGGRTGICCRAARAAYCA) (Leray et al., 2013), widely used for metazoan metabarcoding analysis. The amplification protocol differed between markers because each analysis followed the laboratory workflow previously established and optimized for that marker. Samples were amplified in duplicate for 16S and in triplicate for COI, as broad-range COI metabarcoding can be more sensitive to primer bias and stochastic amplification effects than ribosomal markers such as 16S, particularly in complex eukaryotic communities (Elbrecht et al., 2015). Extraction blanks were included as negative

controls, although they were not sequenced. All PCRs also included negative controls to detect potential contamination. And in no case they yield amplifiable products and were therefore excluded from sequencing.

The 16S PCR protocol consisted of an initial denaturation at 95 °C for 5 min, followed by 28 cycles of 95 °C for 30s, 58 °C for 30s, and 72 °C for 40s, with a final extension at 72 °C for 5 min. Each 16 µl reaction contained 1 µl of DNA template (≤ 20 ng μl^{-1}), 0.5 µl of each primer (10 µM), and 14 µl of 1.1× Red Taq Buffer (Sigma-Aldrich). The concentration of all PCR products was quantified using a Qubit fluorometer. Amplicons were purified with AMPure XP beads (Thermo Fisher Scientific). The duplicate amplicons from each sample were then pooled, and a second PCR (index PCR) of eight cycles under the same conditions was performed using Nextera XT index primers to add unique barcodes and Illumina adapters. Libraries were constructed and normalized to a DNA concentration of 40 ng/µl. 16S libraries were sequenced with paired-end Illumina NextSeq 2000 (2 × 300 bp).

PCR amplification of the COI fragment consisted of an initial denaturation at 95 °C for 10 min, followed by 45 cycles of 95 °C for 45s, 45 °C for 45s, and 72 °C for 45s, and a final extension at 72 °C for 5 min. Each 20 µl reaction contained 1× MyFiTM mix (Meridian Bioscience), 0.08 mg ml^{-1} BSA, and 40 ng of template DNA. Each successful PCR replicate was individually tagged in a second PCR that added UDI barcodes to both ends of the amplicon. Barcoded samples were pooled equimolarly into libraries of up to 96 samples, to which Illumina adapters and library identifiers were added by ligation using the NEXTFLEX Rapid DNA-Seq Kit 2.0 for Illumina (PerkinElmer). The three COI replicates per sample were individually tagged and sequenced separately to reduce the influence of stochastic PCR amplification. The three COI PCR replicates were treated as independent technical replicates during library preparation and sequencing, and were subsequently merged by sample during downstream bioinformatic processing. Libraries were sequenced on an Illumina NextSeq using a V3 600-cycle kit (2 × 300 bp). All sequencing was performed by the Genomics Service at the Universidad Complutense de Madrid.

Bioinformatic sequence processing

The 16S raw reads were first analyzed using FASTQC (Andrews, 2010) and MultiQC (Ewels et al., 2016) to assess overall sequencing quality. All samples showed adequate sequencing metrics, including Phred scores above Q30 for most bases, balanced base composition, and no evidence of adapter contamination. The data were then processed using the QIIME2 2024.10 platform (Bolyen et al., 2019) following an ASV-based workflow, which is well suited to 16S metabarcoding, because the targeted region of the 16S rRNA gene is relatively conserved and supported by curated bacterial reference databases. Primer trimming was performed using the cutadapt plugin (Martin, 2011), allowing a 10% error rate and requiring a minimum overlap of 10 bp. Untrimmed sequences were discarded. Reads were denoised, merged, and filtered using the DADA2 *denoise-paired* function (Callahan et al., 2016). Forward reads were truncated at 180 bp, and reverse reads at 150 bp. A minimum overlap of 20 bp was required, allowing a maximum of two mismatches between forward and reverse reads. Chimeric sequences were removed using the consensus method, which identifies chimeras based on their distribution across samples. After amplification and quality control filtering, 25 samples were retained for the 16S analysis (Table S1).

The COI raw reads were analyzed using a custom pipeline combining denoising and clustering, because COI typically exhibits higher sequence variability, greater intraspecific variation, and less standardized reference coverage across broad eukaryotic groups (Antich et al., 2021). Primer trimming was performed using cutadapt (Martin, 2011). Forward and reverse reads were merged using Flash2 v. 2.2.0 (Magoc & Salzberg, 2011) with a minimum overlap of 10 bp and a maximum of 100 bp. Sequence denoising was performed using UNOISE2, setting a minimum abundance threshold of two reads (*minsize* = 2) to remove singletons, which are more likely to represent sequencing errors, artifacts, or chimeric sequences. Zero-radius amplicon sequence variants (ASVs) representing the same biological entity were clustered into Operational Taxonomic Units (OTUs) to reduce redundancy and facilitate ecological interpretation. Clustering was performed separately for each sample using swarm v. 3.1.5 (Mahé et al., 2022), which clusters sequences based on a snowball-growing approach

that progressively incorporates closely related variants. Up to two differences between neighboring haplotypes were allowed. Replicates with read counts significantly below the mean replicate read count were removed, based on a normal distribution fitted with the *fitdistr* function of the R package MASS. Replicates from the same sample were merged by summing ASV/OTU read counts. The resulting aggregated count tables were subsequently normalized to relative abundances within each sample. One sample with only one replicate and poor-quality metrics was discarded. After filtering, 27 samples were retained for the COI analysis, with technical PCR replicates merged by sample for downstream analyses (Table S1).

Taxonomic assignment

To assign prokaryotic 16S sequences, we first trained a Naïve Bayes classifier using the SILVA 138 99% OTUs reference database (Quast et al., 2012), trimmed to the 515F-806R region. Taxonomy was assigned to the ASVs using the feature-classifier sklearn plugin (Bokulich et al., 2018) in QIIME 2. A confidence threshold of 70% was set, representing an optimal balance between accuracy and coverage (Bokulich et al., 2018). The processed data were imported into an R phyloseq object using the phyloseq package v. 1.48.0 (McMurdie & Holmes, 2013) in R v. 4.4.1 (R Core Team, 2024). To minimize the inclusion of potential sequencing artifacts, only bacterial ASVs represented by at least 10 reads across all samples were retained for subsequent analyses.

Taxonomic assignment of eukaryotic COI reads was performed using the Midori2 (Leray et al., 2022), BOLD (Ratnasingham et al., 2024), and rCRUX (Curd et al., 2024) databases using the SINTAX classifier implemented in VSEARCH v. 2.22.1 (Rognes et al., 2016) and BLAST+. For SINTAX, a minimum confidence threshold of 0.7 was applied. For BLAST+, three identity thresholds were used (99%, 97%, and 95%) with a minimum cutoff of 90%. For each threshold, the last common ancestor (LCA) was determined from sequences showing identity greater than or equal to the corresponding threshold. Species-level assignments were retained only for matches with $\geq 97\%$ identity. Otherwise, taxonomic identification was truncated to genus, family, or higher taxonomic ranks according to the identity obtained. When no

match reached a given threshold, the best available matches with $\geq 90\%$ identity were used to calculate the LCA. This procedure was applied independently to each of the three reference databases (Midori2, BOLD, and rCRUX). Downstream taxonomic and community analyses were restricted to OTUs assigned to at least one taxonomic rank in at least one reference database, in order to focus interpretation on identifiable aquatic lineages. Taxonomic assignments were then manually inspected, and evident misassignments were removed from further analyses, including fully terrestrial organisms not relevant for understanding aquatic biodiversity (i.e., birds, mammals, reptiles, and terrestrial arachnids), and fully marine taxa (i.e., the phyla Echinodermata and Hemichordata, the classes Anthozoa, Scyphozoa, Staurozoa, Cubozoa, Cephalopoda, Chondrichthyes, Thaliacea, and Ascidiacea, and the orders Balanomorpha, Nudibranchia, and Neogastropoda).

A list of taxon occurrence for 16S ASVs and COI OTUs across lakes and habitats with their taxonomic assignment is shown in Supplementary File 1.

Data analysis and diversity measures

Environmental variables were first standardized using z-score transformations to enable comparisons among measurements. Collinearity among variables was assessed using pairwise Spearman's rank correlations calculated with the *cor* function in the base R stats package. Redundant, highly correlated variables ($> 90\%$) representing the same physical feature were reduced to a single variable (Table S3), whereas variables reflecting distinct features were retained even when highly correlated. The remaining non-collinear variables, temperature, DO, DOC, nitrate, phosphate, and salinity, were summarized using Principal component analysis (PCA), performed with the *prcomp* function to identify the main environmental gradients structuring variation among lakes. Spearman's correlations were subsequently used to evaluate relationships between the main PCA axes and lake physical features (area, mean depth, maximum depth, and age). Because only the six focal lakes were available for this analysis, correlations were interpreted cautiously as broad environmental tendencies rather than general lake-level relationships.

Redundancy analysis (RDA) was used to identify the main prokaryotic and eukaryotic phyla

contributing to compositional differences among lakes. Community data from water samples were agglomerated at the phylum level using the *tax_glom* function in phyloseq. The resulting relative-abundance matrices were Hellinger-transformed using the *decostand* function in vegan. Constrained ordinations were then performed by RDA, using lake identity as the explanatory variable (*rda* function in vegan). Analyses were conducted separately for prokaryotic and eukaryotic communities.

To assess prokaryotic and eukaryotic diversity in the lakes, alpha diversity was estimated from ASVs and OTUs, respectively. Effective taxonomic diversity was quantified using Hill numbers for each sample with the *hill_taxa* function from the hillR package (Li, 2018), and was subsequently compared among lakes and habitats (pelagic water and benthic sediment). To ensure comparability across samples and lakes, reads were rarefied to the minimum sequencing depth using the *rrarefy* function in vegan v. 2.6–10 (Oksanen et al., 2025). Rarefaction was performed separately for the COI and 16S datasets. This procedure was repeated 1000 times, and Hill numbers were calculated for each iteration and then averaged for each sample. Hill numbers were calculated for orders $q = 0$, 1, and 2. As q increases, diversity estimates become increasingly weighted toward common and abundant taxa. Hill number $q = 0$ (0D) represents ASV or OTU richness and is insensitive to taxon abundances. Hill number $q = 1$ (1D) represents the effective number of common taxa, and weights taxa according to their relative abundance, analogous to the exponential Shannon's diversity. Hill number $q = 2$ (2D) represents the effective number of dominant taxa, giving greater weight to abundant taxa, analogous to the inverse of Simpson index (Alberdi & Gilbert, 2019). Hill numbers provide a unified framework for comparing diversity by integrating richness and evenness with a common diversity scale (Chao et al., 2010). To test for differences in alpha diversity among lakes, non-parametric Kruskal–Wallis tests were performed using lake identity as the grouping variable. Pairwise post hoc comparisons between lakes were conducted using Dunn's test, as implemented in the FSA package v. 0.10.0 (Ogle et al., 2026). To assess whether alpha diversity varied with depth within lakes, exploratory within-lake analyses were performed. Because sampling depth, sampling year, and replication were not fully balanced across

lakes, these analyses were considered exploratory and interpreted with caution. For prokaryotic communities, for which samples from two depths were available, Wilcoxon rank-sum tests were used. For eukaryotic communities, depth comparisons combined samples from the two sampling campaigns because no single campaign included all depths. Consequently, these analyses should be interpreted as exploratory.

Spearman's correlations were calculated between lake-level mean alpha-diversity metrics and scores of the environmental PCA axes. Alpha-diversity metrics were averaged within each lake to match the scale of the environmental predictors. For each significant lake-level association, a leave-one-out sensitivity analysis was performed by recalculating Spearman's correlations after sequentially excluding each lake.

Differences in prokaryotic and eukaryotic community composition were assessed using beta diversity metrics. Relative abundances were first calculated with the *transform_sample_counts* function in the phyloseq package. Bray–Curtis dissimilarity matrices were then computed for both prokaryotic and eukaryotic communities, whereas unweighted UniFrac distances were calculated for prokaryotic communities only. UniFrac distances incorporate phylogenetic relationships among ASVs and allow assessment of whether bacterial communities are similar both compositionally and phylogenetically. All distance matrices were calculated using the *distance* function in phyloseq. Principal Coordinate Analysis (PCoA) was used to further assess patterns of community dissimilarity among lakes and habitats. Differences among lakes were tested using a permutational multivariate analysis of variance (PERMANOVA) with 100,000 permutations implemented with the *adonis2* function of the vegan package. The assumption of homogeneity of multivariate dispersion among lakes was evaluated using the *betadisper* function in vegan, followed by a permutation test with the *permutest* function and 1000 permutations. Pairwise PERMANOVA was then performed as a post hoc analysis using the *pairwise.adonis* function in vegan to identify pairwise differences in community composition among lakes. To examine community composition at different taxonomic levels, ASVs or OTUs were collapsed using the *psmelt* function in phyloseq. Shared and unique genera and families across lakes were visualized using UpSet diagrams generated with the *upset*

function in UpSetR package v. 1.4.0 (Conway et al., 2017).

Spearman's correlations were calculated between patterns of beta diversity, summarized by PCoA centroids for each lake, environmental PCA axes, and lake physical variables. Centroid values were used to match the lake-level resolution of the environmental data and avoid pseudoreplication. Because only six lakes were available, these analyses were interpreted cautiously as exploratory assessments of broad environmental associations.

Graphics were generated using the ggplot2 package (Wickham, 2016).

Results

Environmental and morphometric characterization of the Nicaraguan lakes

The PCA revealed two major environmental gradients among lakes, with the first two principal components (PCs) explaining 90.3% of the total variation (Fig. 2a). PC1 captured a trophic gradient, with negative scores associated with higher DOC, nitrate, and phosphate concentrations, and positive scores associated with higher dissolved oxygen and temperature. PC2 captured an ionic gradient that differentiated lakes primarily according to salinity (Table S4). The large Nicaraguan lakes, L Nicaragua and L Managua, were shallow and more eutrophic, while the crater lakes were smaller and deeper, warmer, and more

oxygenated. A pronounced salinity gradient was also observed among the crater lakes (Fig. 2a). Lake area and depth, which were not included in the environmental PCA, clearly distinguished the Great Lakes from the crater lakes (Fig. 2b). Additional geomorphometric and historical lake characteristics showed only marginal correlations with the environmental gradients (Table S5). Mean depth was strongly correlated with maximum depth, as expected, but was only marginally associated with the trophic gradient represented by PC1 and was unrelated to the salinity-related PC2 axis. These correlations among lake-level variables should be interpreted cautiously because of the limited number of lakes.

Fish richness was highly uneven among the lakes (Table 1). The Great Lakes harbored substantially more fish species, including cichlids, than the crater lakes, whereas Midas cichlid species richness was highest in the deeper crater lakes, CL Apoyo and CL Xiloá (Table S2; Fig. S1).

Sequencing output and taxonomic assignment

For the prokaryotic 16S dataset, quality filtering, primer removal, and denoising were performed, yielding 5,992,278 retained reads, with an average of 240,743 reads per sample (range: 102,859–355,843). These reads resulted in 22,005 unique amplicon sequence variants (ASVs), assigned to Archaea (2213 ASVs), Bacteria (19,728), Eukarya (5), or no domain (59). After removing non-bacterial and low-abundance ASVs, 12,747 bacterial ASVs were retained

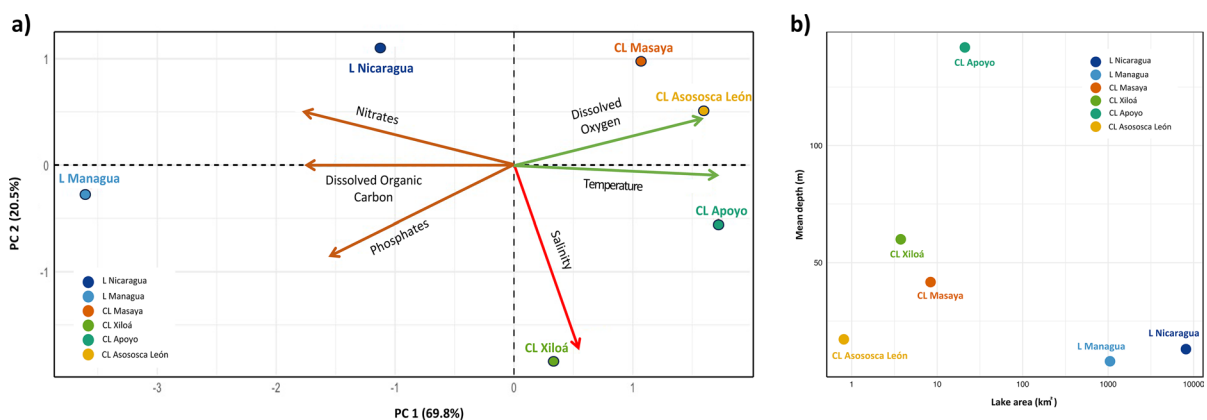


Fig. 2 Environmental and morphometric variation among the Nicaraguan lakes showing **a** a PCA summarizing the main environmental gradients and **b** the relationship between lake size and depth

for downstream analyses. Of these, 6761 (53.04%) were detected exclusively in sediment samples, 3376 (26.48%) exclusively in water samples, and 2610 (20.48%) in both sample types. Taxonomic assignment was successful for 98.14% of ASVs at the phylum level, 97.74% at the class level, 96.06% at the order level, 94.62% at the family level, 88.48% at the genus level, and 58.95% at the species level.

For the eukaryotic COI dataset, primer removal, forward and reverse read merging, and denoising yielded 3,262,517 retained reads, with an average of 98,864 reads per sample (range: 45,066–183,960). These reads clustered into 25,584 OTUs, of which 5141 (20.09%) were successfully assigned to at least one taxonomic level. Among the taxonomically assigned OTUs, assignment success was 96.95% at the phylum level, 92.14% at the class level, 88.10% at the order level, 69.80% at the family level, 65.56% at the genus level, and 49.82% at the species level. Among these 5141 taxonomically assigned OTUs, 1022 (19.88%) were detected exclusively in sediment samples, 3900 (75.86%) exclusively in water samples, and 219 (4.26%) in both sample types. After removing terrestrial taxa, 5082 OTUs remained. Following the additional removal of probable marine taxa, 4917 OTUs were retained for downstream analyses.

Alpha-diversity patterns across lakes and habitats

Standardized alpha-diversity metrics revealed differences in bacterial richness, diversity, and evenness among lakes and between habitats (Table 2;

Fig. 3). For water samples, these differences were statistically supported among lakes (Kruskal–Wallis test: 0D , $\chi^2 = 16.783$, $df = 5$, $P < 0.05$; 1D , $\chi^2 = 18.293$, $df = 5$, $P < 0.05$; 2D , $\chi^2 = 17.488$, $df = 5$, $P < 0.05$). Post hoc analyses showed that significant differences were restricted to a limited number of pairwise comparisons (Table S6). Because only a single sediment sample was available per lake, statistical analyses could not be performed for this habitat.

Alpha-diversity patterns showed higher bacterial diversity in sediment than in water across all lakes, although sediment was represented by a single sample per lake (Table 2; Fig. 3a). Sediment effective diversity was highest in L Nicaragua and lowest in CL Xiloá. In water, L Managua showed markedly higher diversity values than the other lakes. Dominance patterns (2D) also differed between habitats and among lakes. In most lakes, sediment communities were not only more diverse but also more even, whereas water communities showed lower diversity and stronger dominance by a few taxa.

Eukaryotic diversity based on assigned OTUs also differed between lakes and habitats (Table 3; Fig. 3b). Among water samples, differences among lakes were statistically supported for 0D (Kruskal–Wallis test: $\chi^2 = 14.109$, $df = 5$, $P < 0.05$), 1D (Kruskal–Wallis test: $\chi^2 = 15.441$, $df = 5$, $P < 0.05$), and 2D (Kruskal–Wallis test: $\chi^2 = 15.154$, $df = 5$, $P < 0.05$). Post hoc analyses indicated that these differences were significant only for a small number of pairwise comparisons (Table S7). Because sediment was represented by a

Table 2 Bacterial alpha diversity by lake and habitat, expressed as Hill numbers (D) of orders $q = 0, 1$, and 2 , representing ASV richness, and the effective numbers of common and dominant ASVs, respectively

Lake	Water n	Water			Sediment		
		0D	1D	2D	0D	1D	2D
L Nicaragua	2	1141.36	131.26	28.99	2755.81	1396.61	659.42
L Managua	3	1827.07	494.16	128.40	NA	NA	NA
CL Masaya	3	1139.89	165.32	47.26	1925.91	650.31	206.84
CL Xiloá	4	674.27	106.29	46.16	1474.00	343.21	53.88
CL Apoyo	4	661.14	88.45	40.15	3125.00	951.26	310.65
CL Asososca León	4	413.25	56.95	23.28	2223.95	945.53	397.24

Water values represent means across the number of samples indicated in the column n. Sediment values correspond to a single sample per lake, except for L Managua, where no sediment sample was available. Diversity estimates were averaged over 1000 random subsampling iterations

NA No sediment samples available

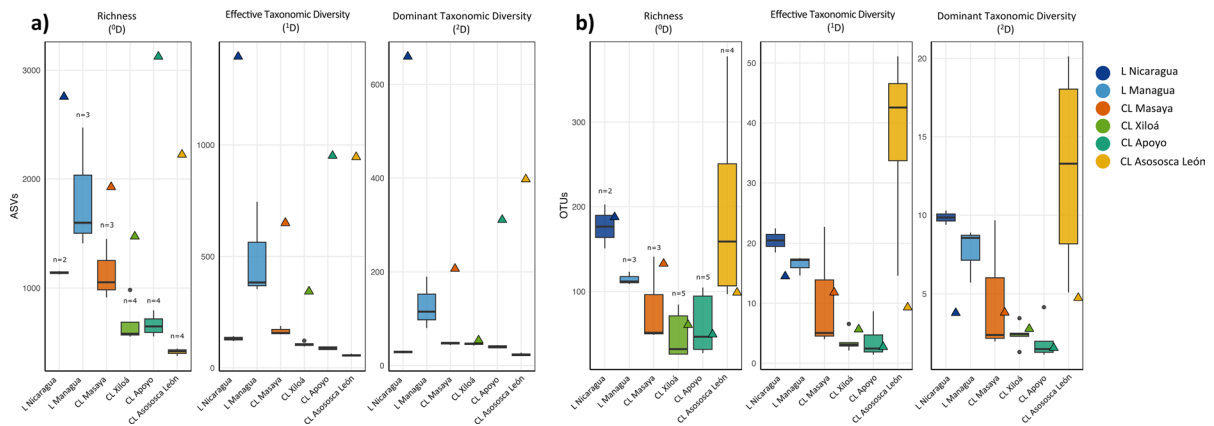


Fig. 3 **a** Bacterial alpha diversity (ASVs) and **b** eukaryotic alpha diversity (OTUs) expressed as Hill numbers of orders $q = 0, 1$, and 2 . Boxplots show water samples, and triangles indicate single sediment samples. The number of water samples

(n) for each lake is shown above the corresponding boxplot in the $q = 0$ panels. Sediment was represented by a single sample per lake, except for L. Managua, where no sediment sample was available

Table 3 Eukaryotic alpha diversity by lake and habitat, expressed as Hill numbers (D) of orders $q = 0, 1$, and 2 , representing ASV richness, and the effective numbers of common and dominant ASVs, respectively

Lake	Water n	Water			Sediment		
		0D	1D	2D	0D	1D	2D
L. Nicaragua	2	176.82	20.49	9.86	187.68	14.46	3.77
L. Managua	3	114.90	16.46	7.73	NA	NA	NA
CL Masaya	3	80.37	10.60	4.67	133.15	11.82	3.80
CL Xiloá	5	47.86	3.62	2.39	60.72	5.37	2.76
CL Apoyo	5	61.04	3.82	2.00	49.49	2.72	1.55
CL Asososca León	4	198.26	37.70	12.94	98.70	9.29	4.74

Water values represent means across the number of samples indicated in the column n after averaging diversity estimates over 1000 random subsampling iterations for each sample. Sediment values correspond to a single sample per lake, except for L. Managua, where no sediment sample was available

NA no sediment samples available

single sample per lake, statistical analyses could not be performed for this habitat.

Sediments generally harbored lower eukaryotic diversity than the water column (mean diversity across water samples within each lake), except in CL Xiloá and CL Masaya, where diversity was similar between habitats (Table 3; Fig. 3b). These comparisons are descriptive because sediment samples were not replicated. Alpha-diversity estimates indicated high community unevenness in both habitats, with only a small subset of OTUs dominating the communities. CL Asososca León showed the highest community diversity and evenness. The lowest

diversity was observed in CL Apoyo and CL Xiloá in both water and sediment samples. The large lakes harbored rich and diverse communities, although these remained highly uneven.

Exploratory within-lake analyses did not reveal consistent differences in alpha diversity with depth. For prokaryotic communities, pairwise comparisons between depths (Table S8) were not significant for Hill numbers $q = 0$, $q = 1$, or $q = 2$. Similarly, for eukaryotic communities, Kruskal–Wallis tests across depths (Table S9) were also not significant. Because sampling depth, sampling year, and

sediment communities appeared to cluster more tightly than water communities, suggesting greater similarity among benthic assemblages across lakes. However, this pattern should be interpreted cautiously because sediment samples were not replicated. In contrast, water communities were more differentiated and showed clearer lake-specific signatures. PCoA ordinations colored by sampling depth (Figs. S4 and S5) did not reveal consistent vertical structuring. Instead, samples clustered primarily according to lake identity, indicating that among-lake differences outweighed depth-related variation. Because sampling depth was uneven among lakes and partially confounded with sampling year, no formal statistical tests of depth effects were performed.

For prokaryotes, PCoA ordinations based on both Bray–Curtis dissimilarity and UniFrac distances showed consistent patterns (Fig. S2). Water communities from the Great Lakes, L Nicaragua and L Managua, clustered together and were closely associated with those of CL Masaya (Fig. S2). In contrast, CL Xiloá and CL Apoyo formed a separate cluster in the UniFrac PCoA, whereas CL Asososca León consistently appeared as the most distinct lake. Dispersion tests confirmed homogeneity of variance among lakes for water samples (Bray–Curtis: $F = 1.344$, $P > 0.05$; UniFrac: $F = 0.864$, $P > 0.05$). PERMANOVA analyses detected significant differences in bacterial community composition among lakes (Bray–Curtis: $R^2 = 0.921$, $F = 32.326$, $P < 0.05$; UniFrac: $R^2 = 0.738$, $F = 7.85$, $P < 0.05$). Pairwise PERMANOVA comparisons indicated that most lakes differed significantly, except for some comparisons involving L Nicaragua (Table S11).

Eukaryotic communities showed broadly similar beta diversity patterns (Fig. S3). Bray–Curtis PCoA revealed that sediment samples clustered together but also tended to group closer to water samples from the same lake, suggesting a lake-specific influence on community composition despite the overall similarity among sediment samples (Fig. S3). Water samples from CL Xiloá and CL Apoyo formed a cluster, indicating similar eukaryotic community composition. As observed for prokaryotes, CL Asososca León occupied a distinct position, although it was closer to the Great Lakes than to CL Xiloá and CL Apoyo. Dispersion tests indicated homogeneous variance among lakes for eukaryotic water communities ($F = 2.377$, $p > 0.05$). PERMANOVA analyses confirmed

significant differences in eukaryotic community composition among lakes (Bray–Curtis: $R^2 = 0.763$, $F = 10.315$, $p < 0.05$). Pairwise comparisons showed significant differences between CL Asososca León and both CL Xiloá and CL Apoyo, between CL Xiloá and both L Managua and CL Masaya, and between CL Apoyo and both L Managua and CL Masaya, while comparisons between CL Asososca León and the two Great Lakes were only marginally significant (Table S12).

To determine whether these compositional differences were associated with lake characteristics, we examined the relationship between the scores on the first PCoA axis (PCoA1) and mean lake depth (Fig. 5). In both prokaryotic and eukaryotic communities, the deepest crater lakes (CL Xiloá and CL Apoyo) fell at one end of PCoA1, whereas the large, shallower lakes fell at the opposite end, suggesting that among-lake variation in community composition was partly structured according to mean depth, rather than simply reflecting the distinction between Great Lakes and crater lakes (Fig. 5). Spearman's correlations between water community composition and the environmental PCA axes were not significant. Nonetheless, several associations were suggestive, particularly between the first environmental principal component (PC1) and the first axis of eukaryotic community composition ($\rho = 0.77$, $p = 0.10$), and between the second environmental principal component (PC2) and the first axis of prokaryotic community composition ($\rho = -0.71$, $p = 0.11$). Mean lake depth, by contrast, was significantly associated with the first axis of eukaryotic community composition ($\rho = 1$, $p = 0.002$) and showed a strong but non-significant association with the first axis of prokaryotic community composition ($\rho = 0.71$, $p = 0.11$; Table S13). These associations reflect differences in mean depth among lakes rather than variation with sampling depth within lakes, as community composition and diversity did not show consistent differences among sampling depths. Given that these analyses were based on six lakes, these correlations should be interpreted as exploratory indicators of broad ecological patterns.

Taxonomic composition of communities

A total of 67 bacterial, 12 archaeal, and 40 eukaryotic phyla were identified across the Nicaraguan lakes,

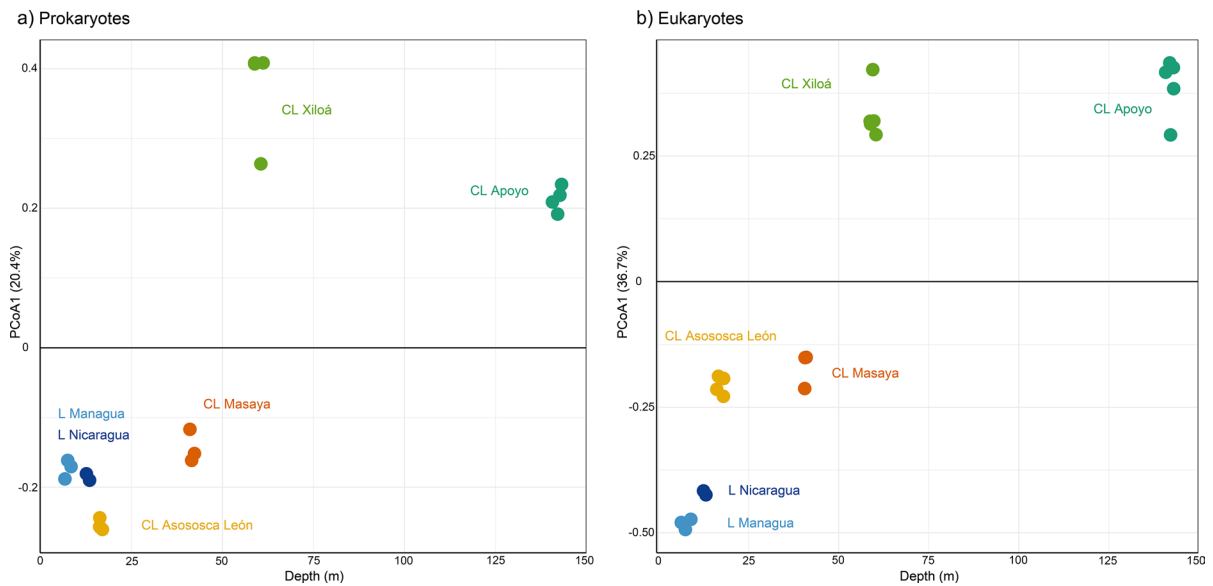


Fig. 5 Relationship between mean lake depth and the first PCoA axis summarizing community structure in water samples for **a** prokaryotes, based on UniFrac distances, and **b** eukaryotes, based on Bray–Curtis dissimilarity

comprising approximately 700 prokaryotic and 700 eukaryotic families. Although most phyla were shared across lakes and habitats, their relative abundances differed markedly between water and sediments and among lakes.

Bacterial community composition at the phylum level in water was strongly structured by lake identity (adjusted $R^2 = 0.787$, $P = 0.001$), with the first two canonical axes of the RDA accounting for 80.0% of the constrained variance. The primary compositional gradient contrasted Actinobacteriota, Cyanobacteria, Verrucomicrobiota, and Bacteroidota with Planctomycetota and Proteobacteria. L Managua was associated with higher relative abundances of Actinobacteriota, whereas L Nicaragua and CL Asososca León were associated with higher relative abundances of Cyanobacteria. In contrast, CL Xiloá and CL Masaya were associated with higher relative abundances of Planctomycetota and Proteobacteria (Fig. S6a). Similarly, eukaryotic community composition was significantly structured by lake identity (adjusted $R^2 = 0.614$, $P = 0.001$), with the first two axes explaining 75.3% of the constrained variance. The main gradient contrasted Chlorophyta with Bacillariophyta, Ochrophyta, and Rotifera, whereas Ascomycota was mainly associated with the second axis. L Managua and CL Masaya were associated with higher relative

abundances of Bacillariophyta, whereas L Nicaragua was associated with higher relative abundances of Arthropoda. CL Asososca León was associated with higher relative abundances of Ascomycota, whereas the deeper lakes, CL Apoyo and CL Xiloá, were associated with higher relative abundances of Chlorophyta and Porifera (Fig. S6b).

Overall, bacterial communities in water were dominated by Cyanobacteria across lakes (Fig. 6a; Fig. S7). Variation among lakes was mainly driven by differences in the relative abundances of the major phyla identified by the RDA, whereas several other phyla, including Chloroflexi, Firmicutes, Acidobacteriota, and Bdellovibrionota, occurred at low relative abundance in most lakes (Fig. 6a; Fig. S7). In sediments, bacterial phylum composition appeared to differ among lakes (Fig. 6b; Fig. S7), although the absence of replicate sediment samples precludes statistical evaluation of these patterns. Proteobacteria, abundant in water, was also abundant in the sediments from all lakes. By contrast, Chloroflexi was rare in water but dominant in lake sediments, representing a prominent component of benthic bacterial communities. Cyanobacteria was highly abundant in CL Masaya and CL Apoyo but less common in the remaining lakes. Firmicutes and Bacteroidota were abundant only in some crater lakes, particularly CL Xiloá, CL Apoyo, and

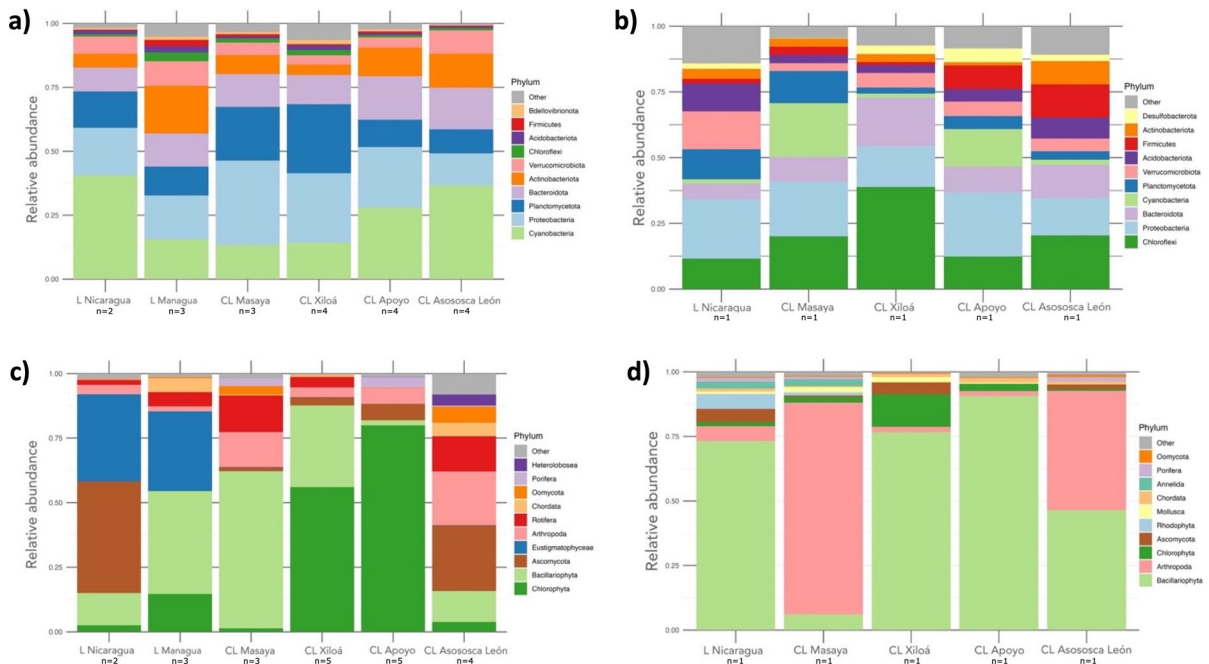


Fig. 6 Relative abundance of the ten most abundant bacterial (**a, b**) and eukaryotic (**c, d**) phyla across lakes in water samples (**a, c**) and sediments (**b, d**). All remaining phyla were grouped as “Other.” The number of water samples (n) is shown below each bar

CL Asososca León, whereas Verrucomicrobiota was especially abundant in L. Nicaragua. Acidobacteriota and Desulfobacterota were also distinctive components of sediment communities. In contrast, Actinobacteriota, highly abundant in water, was scarce in sediments. Several additional low-abundance phyla showed lake-specific distributions. Cyanobacterial communities also differed among lakes in richness and composition, ranging from 35 ASVs and 11 families in CL Asososca León to 184 ASVs and 21 families in L. Managua (Tables S14 and S15). The most common and abundant cyanobacterial family across lakes was Cyanobiaceae, with the highest ASV richness in the large lakes. In CL Masaya, several other cyanobacterial families were also relatively common compared with the other lakes, including Microcystaceae and Nostocaceae, which include taxa frequently associated with bloom-forming cyanobacterial assemblages.

Eukaryotic community composition also differed markedly between habitats and among lakes. Patterns in water broadly matched the RDA results (Fig. 6c; Fig. S8). In lake sediments, Bacillariophyta dominated benthic communities in most lakes, while Arthropoda dominated in CL Masaya and was

also abundant in CL Asososca León (Fig. 6d; Fig. S8). Chlorophyta was less abundant than in water but remained an important component in CL Xiloá. Rhodophyta, a minor component of water communities, was relatively enriched in L. Nicaragua and CL Masaya. Across all samples, 157 arthropod families belonging to 31 orders were detected. L. Nicaragua and CL Asososca León showed the highest arthropod diversity, whereas L. Managua and the saline CL Xiloá showed the lowest. In sediments, Diptera was the most common arthropod group, accounting for up to 40% of reads in all lakes except CL Xiloá and CL Asososca León. In water, the dominant aquatic taxa were calanoid copepods in the crater lakes, Hymenoptera and diplostraca branchiopods in L. Managua, and Diptera in L. Nicaragua and CL Masaya.

Discussion

Environmental heterogeneity of the Nicaraguan lake system

The Nicaraguan lake region, comprising the two Great Lakes and several adjacent isolated crater lakes,

shares a common regional context but does not constitute a homogeneous set of habitats. Rather, the lakes differ markedly in key environmental characteristics. The Great Lakes are larger and hydrologically well connected, both to each other and to surrounding river systems, with L Nicaragua also connected to the Atlantic Ocean through the San Juan River. This connection has allowed the occurrence of marine-associated taxa such as bull sharks and sawfishes (Thorson et al., 1966, 1976; Thorson, 1976a,b, 1982). They are also relatively shallow, bordered by managed land, and exposed to strong winds, promoting frequent mixing throughout the water column. In particular, L Nicaragua is seasonally exposed to northeasterly trade winds, while stronger east-to-west gap-wind events associated with the Papagayo jet occur episodically during the boreal winter (Incer, 1976). This frequent mixing redistributes nutrients from the sediments throughout the water column, supporting elevated and stable plankton biomass also in L Managua (Erikson et al., 1997). In addition, the Great Lakes also exhibit pronounced littoral habitat heterogeneity, with alternating rocky and sandy substrates (Barlow & Munsey, 1976). Together with substantial nutrient inputs, these features are consistent with the high nutrient concentrations and lower dissolved oxygen levels that characterize the more eutrophic conditions of the Great Lakes.

By contrast, the crater lakes are considerably smaller, generally deeper, and often characterized by higher dissolved oxygen concentrations and higher water temperatures, the latter possibly related to their smaller size and ongoing hydrothermal activity and volcanic gas emissions (Villa, 1976). However, they are also highly heterogeneous among themselves. Although these crater lakes originated within a broadly similar and relatively recent time frame, their limnological and colonization histories have not been identical. Some lakes experienced temporary connections to the Great Lakes, for example, CL Xiloá was originally connected to L Managua (Villa, 1968), whereas the historical connections of CL Masaya remain unresolved. By contrast, CL Apoyo and CL Asososca León have remained much more strongly isolated. These differences in connectivity history likely contributed to variation in colonization dynamics and subsequent ecological diversification among lakes. The crater lakes also differ markedly in their physical structure, with two crater lakes being

considerably deeper than the other two, and they span a pronounced salinity gradient that may impose important ecological and selective pressures.

Biodiversity patterns and environmental drivers across habitats and lakes

Our results reveal a broad representation of prokaryotic and eukaryotic diversity across the Nicaraguan lakes, with marked differences between water-column and sediment communities. These differences were reflected in higher prokaryotic diversity in sediments and overall greater eukaryotic diversity in water samples. Water-column communities were dominated by phytoplanktonic primary producers, whereas sediments contained a mixture of aerobic and anaerobic taxa. Fungi and arthropods were abundant in both habitats, while zooplankton, mainly rotifers, was more prominent in the water column, and mollusks were more prevalent in sediments. Consistent with this habitat partitioning, most prokaryotic phyla were found in both habitats, whereas several eukaryotic groups typical of the water column, such as rotifers and eustigmatophytes, were largely absent from sediments. Vertebrates, however, were markedly underrepresented in our eDNA dataset. Only a small fraction of the known ichthyofauna was detected, together with a limited representation of its documented prey and parasites. This likely reflects limitations associated with the COI marker and primer set used in this study rather than the true absence of these taxa. Although COI is widely used for eukaryotic biodiversity assessments, it does not amplify all taxonomic groups with equal efficiency and may underrepresent several non-metazoan eukaryotic lineages. Furthermore, broad-range COI primers may be suboptimal for vertebrate detection because of the high sequence variability of the marker, resulting in reduced or uneven amplification success for fishes and other vertebrates. Primer bias is a well-recognized limitation of metabarcoding studies and can lead to the preferential amplification of some taxa while underrepresenting others (Elbrecht & Leese, 2015; Piñol et al., 2015). Despite these limitations, eDNA sequences were assigned to the bull shark (*Carcharhinus leucas*), a species known to enter L Nicaragua through the San Juan River, illustrating the potential for eDNA approaches to detect rare or unexpected vertebrate taxa.

Beyond habitat differences, community composition also varied strongly among lakes, particularly among crater lakes, suggesting that local environmental conditions contribute to shaping microbial and eukaryotic assemblages across the Nicaraguan lake system. Consistent with the previous studies (Bussing, 1976; Härer et al., 2021), the Great Lakes generally supported high diversity at the sampled sites, although some crater lakes showed similarly or even higher diversity of particular community components. Although L Nicaragua and L Managua harbored the most similar communities overall, they differed in several specific aspects of diversity and taxonomic composition. L Nicaragua, which is larger, deeper, and more hydrologically connected, supported high eukaryotic diversity, consistent with its well-documented fish diversity, as well as particularly high prokaryotic diversity in sediments. Moreover, it was associated with high relative abundances of Cyanobacteria and Arthropoda, a pattern consistent with elevated pelagic productivity and active phytoplankton–zooplankton interactions. In contrast, L Managua, which is more nutrient-rich and potentially more affected by contaminants (Lacayo et al., 1991; McCrary et al., 2006), showed the highest prokaryotic diversity in water samples and was associated with abundant Actinobacteriota, a phylum that includes lineages frequently reported from contaminated environments and capable of tolerating heavy metals and organic pollutants (Behera & Das, 2023). Its association with diatoms (Bacillariophyta) is also consistent with a eutrophic and well-mixed system, where high nutrient availability may sustain elevated primary production. Together, these patterns suggest that trophic state, hydrological connectivity, and water quality may influence specific components of community structure differently in the two Great Lakes, despite their overall composition similarity. This interpretation differs somewhat from that of Härer et al. (2021), who reported similar prokaryotic diversity in L Managua and L Nicaragua. This difference may reflect methodological and temporal variation between studies, including differences in sampling location, sampling design, marker processing, and environmental conditions at the time of collection. Härer et al. (2021) focused on shoreline surface waters, whereas our study included samples from several depths. However, our exploratory analyses did not detect consistent vertical structuring within lakes,

and the difference between studies cannot therefore be attributed specifically to depth. Instead, lake identity remained the dominant source of community differentiation.

The crater lakes, despite their generally lower diversity, were highly heterogeneous. CL Masaya, similarly to L Managua, was dominated by diatoms and was also associated with high relative abundances of Planctomycetota and Proteobacteria, bacterial taxa that include many lineages involved in organic-matter degradation, and their abundance may be consistent with elevated organic inputs from urban runoff (Fuentes Peña, 2019; McCrary, 2024). CL Xiloá and CL Apoyo, the deeper crater lakes, showed the lowest overall diversity and a marked dominance of Chlorophyta. By contrast, the highly isolated CL Asososca León showed the highest eukaryotic diversity, exceeding even that of the Great Lakes, and was associated with high relative abundances of Ascomycota and Cyanobacteria. Several of these patterns were consistent with the taxonomic profiles reported by Härer et al. (2021), including the dominance of Chlorophyta in the water communities of CL Xiloá and CL Apoyo. More generally, several of the dominant bacterial and eukaryotic groups detected in our water samples were also among the most abundant taxa reported by Härer et al. (2021). The recurrence of dominant taxonomic groups across independent studies supports the reproducibility of metabarcoding approaches for characterizing freshwater biological communities and suggests that the major components of these communities may show some temporal persistence at broad taxonomic levels.

Together, these results suggest that isolation, depth, nutrient availability, and other local environmental conditions may interact to shape biodiversity in ways that cannot be explained by lake age and size alone. Although based on limited within-lake replication, the observed patterns are consistent with an important role of local environmental conditions in structuring biological communities across the Nicaraguan lakes.

Our results further corroborate previous work showing that prokaryotic and eukaryotic diversity in the Nicaraguan lakes may respond differently to physical and environmental gradients (Härer et al., 2021). Specifically, prokaryotic richness was positively associated with nutrient availability across lakes, a pattern particularly pronounced in the water

samples from L Managua. This observation is consistent with studies showing that bacterial communities often become richer and more heterogeneous as trophic state increases. Such enrichment may stimulate primary production and increase organic-matter supply, which in turn expands the range of substrates and redox conditions available for microbial metabolism (Jankowski et al., 2014; Kiersztyn et al., 2019; Bakkeren et al., 2025). In contrast, eukaryotic diversity in sediments decreased with mean lake depth and salinity. The association with mean depth may reflect differences among lakes in morphological and environmental characteristics affecting benthic communities, although mean depth itself cannot identify the underlying mechanism. Variation in sediment characteristics, hydrodynamics, littoral influence, or oxygen conditions near the sediment–water interface could potentially contribute to this pattern (Strayer & Findlay, 2010; Poikane et al., 2016; Ntislidou et al., 2018; Wilden et al., 2020). The negative association with salinity further suggests that environmental filtering may restrict the distribution of salt-sensitive taxa, a phenomenon widely documented in freshwater zooplankton and macroinvertebrates (Logares et al., 2013; Kefford et al., 2016). The absence of sensitive arthropod groups such as Heptageniidae mayflies, Leptoceridae caddisflies, and Plecoptera from the relatively saline CL Xiloá is consistent with this interpretation. However, because these associations were evaluated at the lake level using only five lakes for sediments and six lakes for water samples, the correlation analyses should be interpreted as exploratory.

Community differentiation, connectivity, and endemism

The Nicaraguan lakes harbor communities that differ markedly not only in diversity but also in taxonomic composition. The lake-specific distribution of prokaryotic ASVs and eukaryotic OTUs likely reflects the combined effects of ecological differences, environmental filtering, and dispersal limitation within this highly isolated system. In addition, the high levels of endemism documented for several taxonomic groups in the Nicaraguan lakes (Bussing, 1976; Santacruz et al., 2020; Angulo et al., 2023) may further contribute to this differentiation. At broader taxonomic levels, however, more taxa are shared among lakes, indicating high turnover at fine taxonomic resolution but

greater similarity in the major components of community composition.

The Great Lakes harbored the most similar overall community composition, consistent with their historical and still occasional hydrological connection, and CL Masaya also showed strong similarity to them. This pattern agrees with genetic analyses of fish populations indicating a close relationship between Masaya and the Great Lakes (Bracamonte et al., 2022), although whether it reflects recent hydrological connectivity or fish translocation remains unresolved. Interestingly, the two deeper crater lakes also showed strong community similarity despite having no known present or historical connection, suggesting that the ecological conditions associated with deeper lakes may impose similar constraints that favor the assembly of comparable biological communities. Although mean lake depth partly covaried with the broad geomorphological contrast between the shallow Great Lakes and the deeper crater lakes, it did not simply reflect lake type. CL Xiloá and CL Apoyo, the deepest lakes, differed in several other limnological characteristics, suggesting that lake depth captured an ecological dimension of community composition that was not fully explained by the measured environmental variables or by the distinction between crater lakes and Great Lakes. CL Asososca León harbored the most distinct communities, clearly differentiated from those of the other lakes. This marked distinctiveness, together with its higher eukaryotic diversity, makes it a particularly interesting system for studying the evolution of isolated communities. Overall, these results suggest that hydrological connectivity, degree of isolation, and lake morphometry, particularly mean depth, are important correlates of community composition in the Nicaraguan lakes.

Cichlid fish diversity in the Nicaraguan lakes and the diversification of the Midas cichlid

The Nicaraguan lakes are best known for the Midas cichlid species complex, one of the classic examples of rapid adaptive diversification and repeated sympatric evolution in vertebrates (Barluenga et al., 2006; Kautt et al., 2020), and for the bull sharks historically reported from L Nicaragua. These sharks were once thought to represent an endemic landlocked species but were later shown to be bull sharks of Atlantic origin entering through the San

Juan River (Gill & Bransford, 1877; Thorson et al., 1966; Thorson, 1976b). Interestingly, traces of bull shark DNA were also detected in our eDNA survey.

While the Great Lakes harbor a relatively diverse fish fauna, only a few species successfully colonized the crater lakes. Among these, the Midas cichlid underwent extensive diversification following colonization, likely facilitated by ecological opportunity and reduced competition (Kautt et al., 2012, 2020). The deeper crater lakes, particularly Xiloá and Apoyo, repeatedly gave rise to limnetic and benthic ecomorphs adapted to pelagic and deep-benthic habitats (Elmer & Meyer, 2011; Elmer et al., 2014; Kusche et al., 2014; Leal-Cardín et al., 2024). These systems therefore represent well-known examples of parallel evolution. Interestingly, lake depth was also associated with marked differences in community composition across lakes, although eukaryotic sediment diversity declined with increasing mean lake depth.

Our results therefore provide only partial support for the idea that lake characteristics associated with Midas cichlid diversification also structure broader biodiversity patterns. Mean lake depth, which has been implicated in Midas cichlid diversification, was strongly associated with community composition in our eDNA data, but its relationship with diversity did not parallel patterns of cichlid diversification. Conversely, salinity emerged as an important correlate of eukaryotic diversity in sediments despite having no obvious correspondence with Midas cichlid diversity. Thus, the environmental gradients associated with diversification in Midas cichlids overlap only partly with those structuring broader prokaryotic and eukaryotic communities.

Taken together, our results show that the Nicaraguan lake system is structured by a combination of habitat differences, hydrological connectivity, isolation, depth, salinity, and nutrient availability, operating across both ecological and historical timescales. By integrating water and sediment eDNA, this study reveals a broader picture of biodiversity than taxon-specific surveys alone and highlights both shared and distinct environmental associations across different components of biodiversity. The Nicaraguan lakes therefore emerge not as a single limnological unit, but as a mosaic of connected and isolated systems linking community assembly, endemism, and evolutionary diversification.

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Author contributions MB and AR conceived the study. MB and MLC collected the samples. JC performed laboratory work. JC and RG analyzed the data. JC and MB wrote a first draft of the manuscript and all authors contributed to the final version.

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Data availability Raw sequencing data have been deposited to NCBI-SRA archive under the bioproject dataset PRJNA1454747.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval This work has been approved by the Ministerio del Ambiente y los Recursos Naturales of Nicaragua (MARENA).

Informed consent All authors have consented to participate in this paper and approve the submission to Hydrobiologia.

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